

## WMDR2-core

Package in package 'Model'

WMDR2-core

Version 0.1 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-03-11

## WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core

Notes

1. temporalExtent shall be realized as time:interval

2. temporal<Name> elements other than temporalGeometry shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

temporalTerritory = [{

"date": "2000-01-01",

"territory": "<ISO1>"

},

{

"date" 2024-07-13",

"territory": "<ISO2>"

}.

3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

4. Schedules

v0.1.4 Introduce instrument class.

v0.1.5 Expand instrument class.

v0.1.6 introduce more elements

v0.1.8

- require and expanded observation.observedDomain (includes domain, domainFeature [0..1], featureName [0..1])

- introduce temporalGeometry.methods

WMDR2-core

Version 0.1.8

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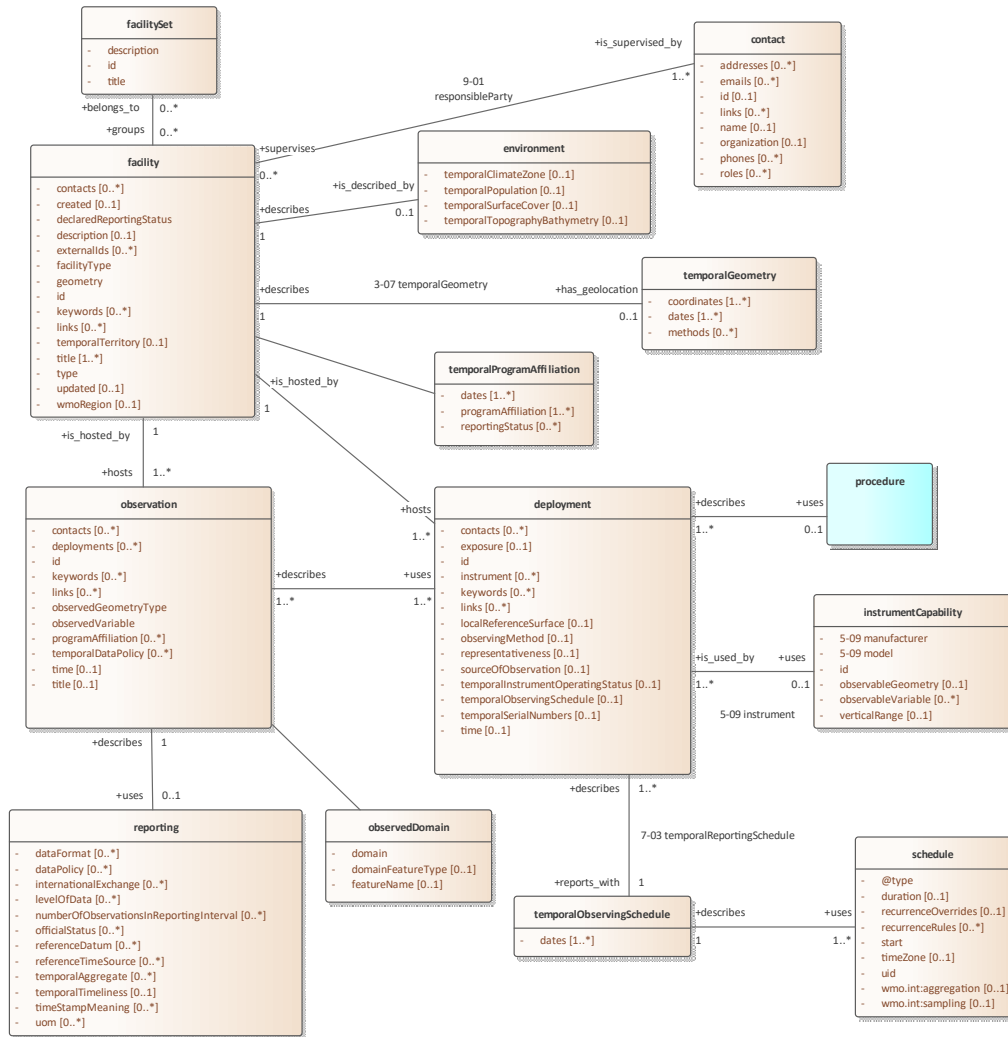


Figure 1: WMDR2-core

## WMDR2-core-logical diagram

Class diagram in package 'WMDR2-core'

WMDR2core

Notes

1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, one for the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

version 0.1.4: simplify observation, specifically specification of 1-01

WMDR2-core-logical

Version 0.1.4

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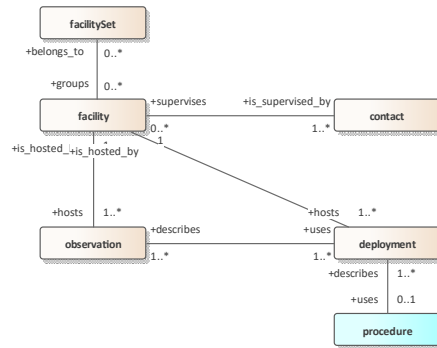


Figure 2: WMDR2-core-logical

## aggregation

Class in package 'WMDR2-core'

WMDR-specific aggregation metadata stored under `schedule["wmo.int:aggregation"]`.

aggregation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                               |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>🔹 diurnalBaseTime : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Optional base time for aggregate alignment, HH:MM:SS.</p> <p>[ Is static True. Containment is Not Specified. ]</p>            |  |
| <p>🔹 spatialResolution : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Optional spatial resolution; prefer numeric metres when known.</p> <p>[ Is static True. Containment is Not Specified. ]</p> |  |
| <p>🔹 statistics : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Optional aggregate statistic values: mean, median, min, max, sum.</p> <p>[ Is static True. Containment is Not Specified. ]</p>     |  |
| <p>🔹 temporalAggregate : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Temporal aggregate period, e.g. PT1H, P1D, P1M.</p> <p>[ Is static True. Containment is Not Specified. ]</p>                |  |

## contact

*Class in package 'WMDR2-core'*

WMDR2 core JSON class: contact.

contact

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                                      |                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <p>◆ addresses : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Postal addresses.</p>                                                                                                                                                      | [ Is static True. Containment is Not Specified. ] |
| <p>◆ emails : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Email contacts.</p>                                                                                                                                                           | [ Is static True. Containment is Not Specified. ] |
| <p>◆ id : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Optional stable contact identifier.</p>                                                                                                                                           | [ Is static True. Containment is Not Specified. ] |
| <p>◆ links : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Online resources associated with the contact.</p>                                                                                                                              | [ Is static True. Containment is Not Specified. ] |
| <p>◆ name : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Either a full name, or a sequence of name parts. Examples:</p> <p>Dr. Jörg Klausen (this corresponds to FN in vcard)<br/>Klausen;Jörg;Dr.; (this corresponds to N in vcard)</p> | [ Is static True. Containment is Not Specified. ] |
| <p>◆ organization : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Example:<br/>ORG:Federal Office of Meteorology and Climatology MeteoSwiss</p>                                                                                           | [ Is static True. Containment is Not Specified. ] |
| <p>◆ phones : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Phone contacts.</p>                                                                                                                                                           | [ Is static True. Containment is Not Specified. ] |

**ATTRIBUTES**

◆ roles : Private  
Multiplicity: ( [0..\*], Allow duplicates: 0, Is ordered: False )

Specific contact roles such as owner, pointOfContact, principalInvestigator.

[ Is static True. Containment is Not Specified. ]

**ASSOCIATIONS**

✎ Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility  
Cardinality: [0..\*]

Target: Public is\_supervised\_by (Class) contact  
Cardinality: [1..\*]

## deployment

Class in package 'WMDR2-core'

WMDR2 core JSON class: deployment.

deployment

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-02

**ATTRIBUTES**

◆ contacts : Private  
Multiplicity: ( [0..\*], Allow duplicates: 0, Is ordered: False )

Deployment-level contacts.

[ Is static True. Containment is Not Specified. ]

◆ exposure : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Compact exposure classification value.

[ Is static True. Containment is Not Specified. ]

◆ id : Private

Deployment identifier; source XML id is preserved when available.

[ Is static True. Containment is Not Specified. ]

◆ instrument : Private  
Multiplicity: ( [0..\*], Allow duplicates: 0, Is ordered: False )

References to reusable instrument ids.

[ Is static True. Containment is Not Specified. ]

◆ keywords : Private  
Multiplicity: ( [0..\*], Allow duplicates: 0, Is ordered: False )

| ATTRIBUTES                                                                                                                                                                                                                                                                                                                              |                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Deployment keywords.                                                                                                                                                                                                                                                                                                                    | [ Is static True. Containment is Not Specified. ] |
|  links : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>Deployment links.                                                                                                                                              | [ Is static True. Containment is Not Specified. ] |
|  localReferenceSurface : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Linked to heightAboveReferenceSurface                                                                                                          | [ Is static True. Containment is Not Specified. ] |
|  observingMethod : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Realized as an array of dictionaries, i.e. key-value pairs:<br>key: observingMethod id<br>values: observation id                                     | [ Is static True. Containment is Not Specified. ] |
|  representativeness : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Compact representativeness value.                                                                                                                 | [ Is static True. Containment is Not Specified. ] |
|  sourceOfObservation : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Compact source-of-observation code value.                                                                                                      | [ Is static True. Containment is Not Specified. ] |
|  temporalInstrumentOperatingStatus : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Wrapper with instrumentOperatingStatus and dates arrays.                                                                         | [ Is static True. Containment is Not Specified. ] |
|  temporalObservingSchedule : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>complex type: when are observations made? Should be expressed as a temporal coverage, e.g. "from Feb-Oct on working days from 06-18 UTC" | [ Is static True. Containment is Not Specified. ] |
|  temporalSerialNumbers : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Implemented as                                                                                                                               |                                                   |

## ATTRIBUTES

```
"temporalInstrumentSerialNumber": {
  "serialNumber": [...],
  "datetimes": [...],
}
```

[ Is static True. Containment is Not Specified. ]

 time : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Deployment temporal extent, date-resolution interval or null.

[ Is static True. Containment is Not Specified. ]

## ASSOCIATIONS

 Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment  
Cardinality: [1..\*]

Target: Public reports\_with (Class)  
temporalObservingSchedule  
Cardinality: [1]

 Association (direction: Unspecified) 5-09 instrument

Source: Public is\_used\_by (Class) deployment  
Cardinality: [1..\*]

Target: Public uses (Class) instrumentCapability  
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public describes (Class) deployment  
Cardinality: [1..\*]

Target: Public uses (Class) procedure  
Cardinality: [0..1]

ObservingProcedure(s)

 Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment  
Cardinality: [0..\*]

Target: Public uses (Class) instrument  
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public is\_hosted\_by (Class) facility  
Cardinality: [1]

Target: Public hosts (Class) deployment  
Cardinality: [1..\*]

 Association (direction: Unspecified)

Source: Public describes (Class) observation  
Cardinality: [1..\*]

Target: Public uses (Class) deployment  
Cardinality: [1..\*]

## environment

*Class in package 'WMDR2-core'*

Optional facility-environment object containing temporal climate, surface-cover, population, and topography/bathymetry histories.

environment

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                                           |                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|  temporalClimateZone : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>Temporal climate-zone history.<br><br>[ Is static True. Containment is Not Specified. ]   |                                                                           |
|  temporalPopulation : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>[ Is static True. Containment is Not Specified. ]                                          |                                                                           |
|  temporalSurfaceCover : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>Temporal surface-cover history.<br><br>[ Is static True. Containment is Not Specified. ] |                                                                           |
|  temporalTopographyBathymetry : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>[ Is static True. Containment is Not Specified. ]                               |                                                                           |
| ASSOCIATIONS                                                                                                                                                                                                                                                                         |                                                                           |
|  Association (direction: Unspecified)<br><br>Source: Public describes (Class) facility<br>Cardinality: [1]                                                                                        | Target: Public is_described_by (Class) environment<br>Cardinality: [0..1] |

**facility***Class in package 'WMDR2-core'*

Facility properties payload of the WMDR2 full record. Root Feature geometry/time are modeled on fullRecord, not here.

facility

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  contacts : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Facility-level contacts. |



| ATTRIBUTES                                                                                                                                                                                           |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|                                                                                                                                                                                                      | [ Is static True. Containment is Not Specified. ] |
| <p>◆ created : Private<br/> Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Record creation date/time metadata.</p>                                                          | [ Is static True. Containment is Not Specified. ] |
| <p>◆ declaredReportingStatus : Private</p>                                                                                                                                                           | [ Is static True. Containment is Not Specified. ] |
| <p>◆ description : Private<br/> Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Facility description.</p>                                                                    | [ Is static True. Containment is Not Specified. ] |
| <p>◆ externalIds : Private<br/> Alias: 3-06 (primary &amp; additional) WSIs<br/> Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>External identifiers such as WMO:WIGOS.</p> | [ Is static True. Containment is Not Specified. ] |
| <p>◆ facilityType : Private<br/> Alias: 3-04 facilityType</p> <p>Compact WMDR facility type code value.</p>                                                                                          | [ Is static True. Containment is Not Specified. ] |
| <p>◆ geometry : Private<br/> Alias: 3-07 current geolocation</p> <p>geometry = geolocation = coordinates + elevation</p>                                                                             | [ Is static True. Containment is Not Specified. ] |
| <p>◆ id : Private<br/> Alias: 3-06 (primary) WSI</p> <p>to prefixed with name space "facility:"</p>                                                                                                  | [ Is static True. Containment is Not Specified. ] |
| <p>◆ keywords : Private<br/> Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Discovery/search keywords retained in the core JSON.</p>                                        | [ Is static True. Containment is Not Specified. ] |
| <p>◆ links : Private<br/> Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p>                                                                                                       |                                                   |

| ATTRIBUTES                                                                                                                                                                                                                                                                                |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Related links.                                                                                                                                                                                                                                                                            | [ Is static True. Containment is Not Specified. ] |
|  temporalTerritory : Private<br>Alias: 3-02 temporal Territory<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>mandatory for fixed facilities, optional for mobile facilities | [ Is static True. Containment is Not Specified. ] |
|  title : Private<br>Alias: 3-03 facilityName<br>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )<br><br>properties.title = 3-03 facilityName                                             | [ Is static True. Containment is Not Specified. ] |
|  type : Private<br><br>Facility properties type; const facility.                                                                                                                                         | [ Is static True. Containment is Not Specified. ] |
|  updated : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>Record update date/time metadata.                                                                          | [ Is static True. Containment is Not Specified. ] |
|  wmoRegion : Private<br>Alias: 3-01 wmoRegion<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>mandatory for fixed facilities, optional for mobile facilities.               | [ Is static True. Containment is Not Specified. ] |

| ASSOCIATIONS                                                                                                                                                                                                        |                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  Association (direction: Unspecified) 3-07 temporalGeometry<br><br>Source: Public describes (Class) facility<br>Cardinality: [1] | Target: Public has_geolocation (Class)<br>temporalGeometry<br>Cardinality: [0..1] |
|  Association (direction: Unspecified)<br><br>Source: Public is_hosted_by (Class) facility<br>Cardinality: [1]                    | Target: Public hosts (Class) deployment<br>Cardinality: [1..*]                    |
|  Association (direction: Unspecified)<br><br>Source: Public groups (Class) facility<br>Cardinality: [0..*]                       | Target: Public belongs_to (Class) facilitySet<br>Cardinality: [0..*]              |

| ASSOCIATIONS                                                                                                                                 |                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|  Association (direction: Unspecified) 9-01 responsibleParty |                                                                           |
| Source: Public supervises (Class) facility<br>Cardinality: [0..*]                                                                            | Target: Public is_supervised_by (Class) contact<br>Cardinality: [1..*]    |
|  Association (direction: Unspecified)                       |                                                                           |
| Source: Public (Class) facility                                                                                                              | Target: Public (Class) temporalProgramAffiliation                         |
|  Association (direction: Unspecified)                       |                                                                           |
| Source: Public is_hosted_by (Class) facility<br>Cardinality: [1]                                                                             | Target: Public hosts (Class) observation<br>Cardinality: [1..*]           |
|  Association (direction: Unspecified)                       |                                                                           |
| Source: Public describes (Class) facility<br>Cardinality: [1]                                                                                | Target: Public is_described_by (Class) environment<br>Cardinality: [0..1] |

## facilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

facilitySet  
Version 1.0 Phase 1.0 Proposed  
jkl created on 2026-03-06. Last modified 2026-06-02

| ATTRIBUTES                                                                                                               |                                                                      |
|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|  description : Private                | [ Is static True. Containment is Not Specified. ]                    |
|  id : Private                         | [ Is static True. Containment is Not Specified. ]                    |
|  title : Private                      | [ Is static True. Containment is Not Specified. ]                    |
| ASSOCIATIONS                                                                                                             |                                                                      |
|  Association (direction: Unspecified) |                                                                      |
| Source: Public groups (Class) facility<br>Cardinality: [0..*]                                                            | Target: Public belongs_to (Class) facilitySet<br>Cardinality: [0..*] |

## fullRecord

Class in package 'WMDR2-core'

Root WMDR2 full-record Feature. This class represents the GeoJSON/OGC Records envelope around the facility-centric WMDR2 payload.

fullRecord

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                                                                         |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  conformsTo : Private<br>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )<br><br>Required conformance URIs including OGC Records core and WMDR2 full-record.<br>[ Is static True. Containment is Not Specified. ] |  |
|  geometry : Private<br><br>Current GeoJSON geometry. Derived from the most recent temporalGeometry coordinate when available; may be null in JSON.<br>[ Is static True. Containment is Not Specified. ]                           |  |
|  id : Private<br><br>Facility full-record identifier, e.g. facility:<WIGOS identifier>.<br>[ Is static True. Containment is Not Specified. ]                                                                                      |  |
|  properties : Private<br><br>Facility properties payload.<br>[ Is static True. Containment is Not Specified. ]                                                                                                                  |  |
|  temporalGeometry : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>WMDR2 facility coordinate history, kept at the root Feature level.<br>[ Is static True. Containment is Not Specified. ]  |  |
|  time : Private<br><br>OGC Records temporal extent of the facility resource. Date-resolution interval or null.<br>[ Is static True. Containment is Not Specified. ]                                                             |  |
|  type : Private<br><br>GeoJSON Feature type; const Feature.<br>[ Is static True. Containment is Not Specified. ]                                                                                                                |  |

## instrument

Class in package 'WMDR2-core'

WMDR2 core JSON class: instrument.

instrument  
Version 1.0 Phase 1.0 Proposed  
jkl created on 2026-03-11. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                                                                           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  id : Private<br>Reusable instrument id.<br>[ Is static True. Containment is Not Specified. ]                                                                                                                                       |  |
|  manufacturer : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>This should come from a registry/catalogue, but should accept any organization.<br>[ Is static True. Containment is Not Specified. ] |  |
|  model : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>This should come from a registry/catalogue, but can accept "custom-built".<br>[ Is static True. Containment is Not Specified. ]             |  |
| ASSOCIATIONS                                                                                                                                                                                                                                                                                                         |  |
|  Association (direction: Unspecified) 5-09 instrument<br>Source: Public describes (Class) deployment<br>Cardinality: [0..*]<br>Target: Public uses (Class) instrument<br>Cardinality: [0..1]                                      |  |

## instrumentCapability

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

instrumentCapability  
Version 1.0 Phase 1.0 Proposed  
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| ATTRIBUTES                                                                                                                                                           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  5-09 manufacturer : Private<br>[ Is static True. Containment is Not Specified. ] |  |
|  5-09 model : Private<br>[ Is static True. Containment is Not Specified. ]        |  |

| ATTRIBUTES                                                                                                                                                                                                                                                                                                                                                             |                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |
|  <b>id</b> : Private                                                                                                                                                                                                                                                                  | [ Is static True. Containment is Not Specified. ]                              |
|                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |
|  <b>observableGeometry</b> : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Specify using <a href="http://codes.wmo.int/wmdr/Geometry">http://codes.wmo.int/wmdr/Geometry</a>                                                                         | [ Is static True. Containment is Not Specified. ]                              |
|                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |
|  <b>observableVariable</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>Specify as an array of values taken from <a href="http://codes.wmo.int/wmdr/ObservedVariable">http://codes.wmo.int/wmdr/ObservedVariable</a> or as free text description. | [ Is static True. Containment is Not Specified. ]                              |
|                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |
|  <b>verticalRange</b> : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.           | [ Is static True. Containment is Not Specified. ]                              |
| ASSOCIATIONS                                                                                                                                                                                                                                                                                                                                                           |                                                                                |
|  Association (direction: Unspecified) 5-09 instrument                                                                                                                                                                                                                               |                                                                                |
| Source: <b>Public is _used_by</b> (Class) deployment<br>Cardinality: [1..*]                                                                                                                                                                                                                                                                                            | Target: <b>Public uses</b> (Class) instrumentCapability<br>Cardinality: [0..1] |

## observation

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

observation  
Version 1.0 Phase 1.0 Proposed  
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| ATTRIBUTES                                                                                                                                                                                                       |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|                                                                                                                                                                                                                  |                                                   |
|  <b>contacts</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>Observation-level contacts. | [ Is static True. Containment is Not Specified. ] |
|                                                                                                                                                                                                                  |                                                   |
|  <b>deployments</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )                             |                                                   |

| ATTRIBUTES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Deployment id references.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [ Is static True. Containment is Not Specified. ] |
|  id : Private<br>Observation identifier, e.g. observation:<observedVariable>.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [ Is static True. Containment is Not Specified. ] |
|  keywords : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>Observation keywords.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | [ Is static True. Containment is Not Specified. ] |
|  links : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>Observation links.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [ Is static True. Containment is Not Specified. ] |
|  observedGeometryType : Private<br>cf. <a href="http://codes.wmo.int/wmdr/Geometry">http://codes.wmo.int/wmdr/Geometry</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [ Is static True. Containment is Not Specified. ] |
|  observedVariable : Private<br><p>The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at <a href="http://codes.wmo.int/wmdr2">codes.wmo.int/wmdr2</a>.</p> <p>NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific.</p> <p>Examples include:</p> <ul style="list-style-type: none"> <li>- observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...)</li> <li>- objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll</li> <li>- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.</li> <li>- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ...</li> </ul> <p>These elements will be supported by code lists from <a href="http://codes.wmo.int/wmdr2">codes.wmo.int/wmdr2</a>.</p> <p>Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.</p> | [ Is static True. Containment is Not Specified. ] |
|  programAffiliation : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>Program-affiliation timeline fragments inherited from WMDR1 observations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | [ Is static True. Containment is Not Specified. ] |

## ATTRIBUTES

 temporalDataPolicy : Private  
Multiplicity: ( [0..\*], Allow duplicates: 0, Is ordered: False )

Optional temporal data-policy history.

[ Is static True. Containment is Not Specified. ]

 time : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Observation temporal extent, date-resolution interval or null.

[ Is static True. Containment is Not Specified. ]

 title : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Human-readable title derived from domain, geometry, observed variable, and optional label.

[ Is static True. Containment is Not Specified. ]

## ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) observation

Target: Public (Class) observedDomain

 Association (direction: Unspecified)

Source: Public describes (Class) observation  
Cardinality: [1..\*]

Target: Public uses (Class) deployment  
Cardinality: [1..\*]

 Association (direction: Unspecified)

Source: Public describes (Class) observation  
Cardinality: [1]

Target: Public uses (Class) reporting  
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public is\_hosted\_by (Class) facility  
Cardinality: [1]

Target: Public hosts (Class) observation  
Cardinality: [1..\*]

## observedDomain

*Class in package 'WMDR2-core'*

observedDomain

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-10. Last modified 2026-06-10

## ATTRIBUTES



## ATTRIBUTES

domain : Private

cf. <http://codes.wmo.int/wmdr/Domain>

[ Is static True. Containment is Not Specified. ]

domainFeatureType : Private

Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Concept motivated by hydrology, but may be useful elsewhere. Propose new code list, with terms like

- river basin
- lake
- river
- catchment
- canopy
- boundary layer
- thermocline
- ...

[ Is static True. Containment is Not Specified. ]

featureName : Private

Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

free text field, e.g., "Danube Basin"

[ Is static True. Containment is Not Specified. ]

## ASSOCIATIONS

Association (direction: Unspecified)

Source: Public (Class) observation

Target: Public (Class) observedDomain

## procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

procedure  
Version 1.0 Phase 1.0 Proposed  
jkl created on 2026-03-11. Last modified 2026-06-02

## ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment  
Cardinality: [1..\*]

Target: Public uses (Class) procedure  
Cardinality: [0..1]

ObservingProcedure(s)

## recurrenceRule

Class in package 'WMDR2-core'

WMDR2 core JSON class: recurrenceRule.

recurrenceRule

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                        |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  @type : Private<br>JSCalendar RecurrenceRule type; const RecurrenceRule.<br>[ Is static True. Containment is Not Specified. ]                                                   |  |
|  byDay : Private<br>Multiplicity: ( [ 0..* ], Allow duplicates: 0, Is ordered: False )<br>Optional day constraints.<br>[ Is static True. Containment is Not Specified. ]         |  |
|  byHour : Private<br>Multiplicity: ( [ 0..* ], Allow duplicates: 0, Is ordered: False )<br>Optional hour constraints.<br>[ Is static True. Containment is Not Specified. ]       |  |
|  byMinute : Private<br>Multiplicity: ( [ 0..* ], Allow duplicates: 0, Is ordered: False )<br>Optional minute constraints.<br>[ Is static True. Containment is Not Specified. ] |  |
|  byMonth : Private<br>Multiplicity: ( [ 0..* ], Allow duplicates: 0, Is ordered: False )<br>Optional month constraints.<br>[ Is static True. Containment is Not Specified. ]   |  |
|  count : Private<br>Multiplicity: ( [ 0..1 ], Allow duplicates: 0, Is ordered: False )<br>Optional recurrence count.<br>[ Is static True. Containment is Not Specified. ]      |  |
|  frequency : Private<br>JSCalendar recurrence frequency.<br>[ Is static True. Containment is Not Specified. ]                                                                  |  |
|                                                                                                                                                                                                                                                                   |  |

| ATTRIBUTES                                                                                                                          |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <p>interval : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Optional recurrence interval.</p> | [ Is static True. Containment is Not Specified. ] |
| <p>until : Private<br/>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )</p> <p>Optional recurrence end.</p>         | [ Is static True. Containment is Not Specified. ] |

## reporting

Class in package 'WMDR2-core'

Observation reporting metadata represented as aligned arrays. temporalTimeliness is itself a temporal wrapper.

reporting  
Version 1.0 Phase 1.0 Proposed  
ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                           |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <p>dataFormat : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Aligned array of compact data-format values.</p>                                                 | [ Is static True. Containment is Not Specified. ] |
| <p>dataPolicy : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Aligned array of data-policy objects, preserving attribution when present.</p>                   | [ Is static True. Containment is Not Specified. ] |
| <p>internationalExchange : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Aligned array of international-exchange flags.</p>                                    | [ Is static True. Containment is Not Specified. ] |
| <p>levelOfData : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Aligned array of compact level-of-data values.</p>                                              | [ Is static True. Containment is Not Specified. ] |
| <p>numberOfObservationsInReportingInterval : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>Aligned array copied from WMDR reporting metadata when present.</p> |                                                   |

| ATTRIBUTES                                                                                                                                                                                                                                                                              |                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                         | [ Is static True. Containment is Not Specified. ]            |
|  <b>officialStatus</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Aligned array of official-status flags.                                                    | [ Is static True. Containment is Not Specified. ]            |
|  <b>referenceDatum</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Aligned array of reference-datum objects.                                                  | [ Is static True. Containment is Not Specified. ]            |
|  <b>referenceTimeSource</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Aligned array of reference-time-source values.                                        | [ Is static True. Containment is Not Specified. ]            |
|  <b>temporalAggregate</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Aligned array of reporting/aggregation periods, renamed from temporalReportingInterval. | [ Is static True. Containment is Not Specified. ]            |
|  <b>temporalTimeliness</b> : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br><br>Optional wrapper with timeliness durations and dates.                                | [ Is static True. Containment is Not Specified. ]            |
|  <b>timeStampMeaning</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Aligned array of timestamp-meaning values.                                             | [ Is static True. Containment is Not Specified. ]            |
|  <b>uom</b> : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br><br>Aligned array of compact unit code values.                                                          | [ Is static True. Containment is Not Specified. ]            |
| ASSOCIATIONS                                                                                                                                                                                                                                                                            |                                                              |
|  Association (direction: Unspecified)                                                                                                                                                                |                                                              |
| Source: Public describes (Class) observation<br>Cardinality: [1]                                                                                                                                                                                                                        | Target: Public uses (Class) reporting<br>Cardinality: [0..1] |

## schedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:\* properties.

schedule  
Version 1.0 Phase 1.0 Proposed  
ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                       |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|  @type : Private<br>JSCalendar object type; const Event.                                                                                                                        | [ Is static True. Containment is Not Specified. ] |
|  duration : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>Occurrence duration describing coverage window or sampling duration.                 | [ Is static True. Containment is Not Specified. ] |
|  recurrenceOverrides : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>JSCalendar recurrence override patches; null override values are invalid. | [ Is static True. Containment is Not Specified. ] |
|  recurrenceRules : Private<br>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )<br>JSCalendar recurrence rules.                                                | [ Is static True. Containment is Not Specified. ] |
|  start : Private<br>Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.                                                                    | [ Is static True. Containment is Not Specified. ] |
|  timeZone : Private<br>Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )<br>IANA time zone or UTC default.                                                     | [ Is static True. Containment is Not Specified. ] |
|  uid : Private<br>JSCalendar-safe schedule uid.                                                                                                                               | [ Is static True. Containment is Not Specified. ] |

**ATTRIBUTES**

🔹 wmo.int:aggregation : Private  
 Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

WMDR-specific aggregation metadata.

[ Is static True. Containment is Not Specified. ]

🔹 wmo.int:sampling : Private  
 Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

WMDR-specific sampling metadata preserved from dataGeneration.sampling.

[ Is static True. Containment is Not Specified. ]

**ASSOCIATIONS**

✎ Association (direction: Unspecified)

Source: Public describes (Class) temporalObservingSchedule  
 Cardinality: [1]

Target: Public uses (Class) schedule  
 Cardinality: [1..\*]

## temporalClimateZone

*Class in package 'WMDR2-core'*

WMDR2 core JSON class: temporalClimateZone.

temporalClimateZone

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

**ATTRIBUTES**

🔹 climateZone : Private  
 Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Compact climate-zone values.

[ Is static True. Containment is Not Specified. ]

🔹 dates : Private  
 Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Date array aligned with climateZone.

[ Is static True. Containment is Not Specified. ]

## temporalGeometry

*Class in package 'WMDR2-core'*

WMDR2 core JSON class: temporalGeometry.

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                                                                    |                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <p>coordinates : Private<br/>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )</p> <p>Coordinate history array; each entry is lon, lat[, elevation].</p> <p>[ Is static True. Containment is Not Specified. ]</p>                                              |                                                                                            |
| <p>dates : Private<br/>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )</p> <p>Date array aligned with coordinates.</p> <p>[ Is static True. Containment is Not Specified. ]</p>                                                                              |                                                                                            |
| <p>methods : Private<br/>Multiplicity: ( [0..*], Allow duplicates: 0, Is ordered: False )</p> <p>use <a href="http://codes.wmo.int/wmdr/GeopositioningMethod">http://codes.wmo.int/wmdr/GeopositioningMethod</a></p> <p>[ Is static True. Containment is Not Specified. ]</p> |                                                                                            |
| ASSOCIATIONS                                                                                                                                                                                                                                                                  |                                                                                            |
| <p>Association (direction: Unspecified) 3-07 temporalGeometry</p> <p>Source: Public describes (Class) facility<br/>Cardinality: [1]</p>                                                                                                                                       | <p>Target: Public has_geolocation (Class)<br/>temporalGeometry<br/>Cardinality: [0..1]</p> |

## temporalInstrumentOperatingStatus

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalInstrumentOperatingStatus.

temporalInstrumentOperatingStatus

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                                                                  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>dates : Private<br/>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )</p> <p>Date array aligned with instrumentOperatingStatus.</p> <p>[ Is static True. Containment is Not Specified. ]</p>              |  |
| <p>instrumentOperatingStatus : Private<br/>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )</p> <p>Compact instrument-operating-status values.</p> <p>[ Is static True. Containment is Not Specified. ]</p> |  |

## temporalObservingSchedule

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalObservingSchedule.

temporalObservingSchedule

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-02

### ATTRIBUTES

🔑 **dates** : Private  
Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Date array aligned with observingSchedule.

[ Is static True. Containment is Not Specified. ]

### ASSOCIATIONS

✎ Association (direction: Unspecified)

Source: Public describes (Class) temporalObservingSchedule  
Cardinality: [1]

Target: Public uses (Class) schedule  
Cardinality: [1..\*]

✎ Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment  
Cardinality: [1..\*]

Target: Public reports\_with (Class)  
temporalObservingSchedule  
Cardinality: [1]

## temporalPopulation

Class in package 'WMDR2-core'

temporalPopulation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

### ATTRIBUTES

🔑 **dates** : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Begin dates aligned with the corresponding values.

[ Is static True. Containment is Not Specified. ]

🔑 **populationDensity** : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Aligned population-density values. The corresponding begin dates are in dates[].

[ Is static True. Containment is Not Specified. ]



## ATTRIBUTES

## temporalProgramAffiliation

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalProgramAffiliation.

temporalProgramAffiliation  
Version 1.0 Phase 1.0 Proposed  
ChatGPT created on 2026-06-01. Last modified 2026-06-02

## ATTRIBUTES

🔑 dates : Private  
Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Date array aligned with programAffiliation.

[ Is static True. Containment is Not Specified. ]

🔑 programAffiliation : Private  
Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Compact program-affiliation values.

[ Is static True. Containment is Not Specified. ]

🔑 reportingStatus : Private  
Multiplicity: ( [0..\*], Allow duplicates: 0, Is ordered: False )

Optional compact reporting-status values aligned with programAffiliation.

[ Is static True. Containment is Not Specified. ]

## ASSOCIATIONS

✎ Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (Class) temporalProgramAffiliation

## temporalSerialNumbers

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalSerialNumbers.

temporalSerialNumbers  
Version 1.0 Phase 1.0 Proposed  
ChatGPT created on 2026-06-01. Last modified 2026-06-02

## ATTRIBUTES

🔑 dates : Private

| ATTRIBUTES                                                                                                                                                                   |                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )                                                                                                             |                                                   |
| Date array aligned with serialNumber.                                                                                                                                        | [ Is static True. Containment is Not Specified. ] |
|                                                                                                                                                                              |                                                   |
|  serialNumber : Private<br>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False ) |                                                   |
| Serial-number history values.                                                                                                                                                | [ Is static True. Containment is Not Specified. ] |

## temporalSurfaceCover

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalSurfaceCover.

temporalSurfaceCover  
Version 1.0 Phase 1.0 Proposed  
ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                                     |                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|  dates : Private<br>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False )          |                                                   |
| Date array aligned with surfaceCover.                                                                                                                                          | [ Is static True. Containment is Not Specified. ] |
|                                                                                                                                                                                |                                                   |
|  surfaceCover : Private<br>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False ) |                                                   |
| Compact surface-cover values.                                                                                                                                                  | [ Is static True. Containment is Not Specified. ] |

## temporalTerritory

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalTerritory.

temporalTerritory  
Version 1.0 Phase 1.0 Proposed  
ChatGPT created on 2026-06-01. Last modified 2026-06-02

| ATTRIBUTES                                                                                                                                                              |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
|  dates : Private<br>Multiplicity: ( [1..*], Allow duplicates: 0, Is ordered: False ) |                                                   |
| Date array aligned with territory.                                                                                                                                      | [ Is static True. Containment is Not Specified. ] |

## ATTRIBUTES

territory : Private  
Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Compact territory values.

[ Is static True. Containment is Not Specified. ]

## temporalTimeliness

Class in package 'WMDR2-core'

Temporal wrapper for reporting latency/delay durations.

temporalTimeliness

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

## ATTRIBUTES

dates : Private  
Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Date array aligned with timeliness.

[ Is static True. Containment is Not Specified. ]

timeliness : Private  
Multiplicity: ( [1..\*], Allow duplicates: 0, Is ordered: False )

Reporting latency/delay durations.

[ Is static True. Containment is Not Specified. ]

## temporalTopographyBathymetry

Class in package 'WMDR2-core'

temporalTopographyBathymetry

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

## ATTRIBUTES

dates : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Begin dates aligned with the corresponding values.

[ Is static True. Containment is Not Specified. ]

topographyBathymetry : Private  
Multiplicity: ( [0..1], Allow duplicates: 0, Is ordered: False )

Topography/bathymetry values using the elements available in the WMDR source XML.

| ATTRIBUTES |                                                   |
|------------|---------------------------------------------------|
|            | [ Is static True. Containment is Not Specified. ] |